

Homework 2: Bandits

CS/Stat 184(0): Introduction to Reinforcement Learning

(Due October 13, 2024 at 11:59 pm ET)

0 Instructions

For each question in this HW, please list all your collaborators and reference materials (beyond those specified on the website) that were used for this homework. Please add your remarks in a “Question 0”.

Submission format: 1) Submit your report as a single pdf file on Gradescope under the assignment “HW2” only. Please tag your pages correctly as is requested in Gradescope. We highly encourage you to use LaTeX. We will also accept neatly written handwritten homework. 2) Please see the separate programming assignment for submission instructions there.

1 Bias of $\hat{\mu}_t^{(k)}$

In the multi-armed bandit setting, although rewards from a given arm are i.i.d., the sample mean of the rewards from a given arm k up to a given timestep t , that is,

$$\hat{\mu}_t^{(k)} := \frac{1}{N_t^{(k)}} \sum_{\tau=0}^{t-1} 1_{\{a_\tau=k\}} r_\tau,$$

does not have the same nice properties as the sample mean of i.i.d. samples.

In this problem, we will see that $\hat{\mu}_t^{(k)}$ is a biased estimator of $\mu^{(k)}$, that is,

$$\mathbb{E}[\hat{\mu}_t^{(k)}] - \mu^{(k)} \neq 0.$$

Consider a bandit with just $K = 2$ arms and a time horizon of $T = 3$. Any reasonable algorithm should sample each arm once, so we assume $a_0 = 1$ and $a_1 = 2$, which leaves the choice of a_2 . We choose the greedy action, i.e.,

$$a_2 = \begin{cases} 1 & \text{if } r_0 \geq r_1 \\ 2 & \text{if } r_0 < r_1 \end{cases}$$

Hint: It may be useful to recall the law of total expectation: for a random variable X and an event A (with complement A^c),

$$\mathbb{E}[X] = \mathbb{E}[X \mid A] \mathbb{P}(A) + \mathbb{E}[X \mid A^c] \mathbb{P}(A^c).$$

For the next two parts, suppose $\nu^{(1)} = \nu^{(2)} = \nu$, i.e., returns from arms 1 and 2 follow the same distribution ν , and that ν has a continuous distribution (i.e., $\mathbb{P}_{r \sim \nu}(r = x) = 0$ for all x), $\mathbb{E}_{r \sim \nu}[r] = 0$, and furthermore ν is symmetrical around zero (i.e., $\mathbb{P}_{r \sim \nu}(r \leq -x) = \mathbb{P}_{r \sim \nu}(r \geq x)$).

1. [4 points] Show that $\mathbb{E}[r_0 \mid r_0 < r_1] = -\mathbb{E}[r_0 \mid r_0 > r_1]$.
2. [6 points] Now show that $\mathbb{E}[\hat{\mu}_3^{(1)}] < 0$ (which by symmetry implies $\mathbb{E}[\hat{\mu}_3^{(2)}] < 0$).
3. [10 points] Now suppose $\nu^{(1)} = \text{Uniform}(0, 1)$ and $\nu^{(2)}$ is the distribution which places probability 1 on the value a for some $a \in [0, 1]$ (i.e., $\mathbb{P}_{r \sim \nu^{(2)}}(r = a) = 1$). Compute $\mathbb{E}[\hat{\mu}_3^{(1)}]$ as a function of a . For which values of a is $\hat{\mu}_3^{(1)}$ biased upwards, downwards, and unbiased?

2 ε -greedy algorithm

Consider the Bernoulli bandit setting where arm k 's rewards follow a Bernoulli distribution with mean (i.e. success probability) $\mu^{(k)} \in [0, 1]$. Recall that $\text{Regret}_T = \sum_{t=0}^{T-1} \mu^{(k^*)} - \mu^{(a_t)}$, and hence the expected regret is

$$\mathbb{E}[\text{Regret}_T] = \sum_{t=0}^{T-1} \mathbb{E}[\mu^{(k^*)} - \mu^{(a_t)}].$$

1. [8 points] Recall $N_t^{(k)} = \sum_{\tau=0}^{t-1} 1_{\{a_\tau=k\}}$ is the number of times arm k is pulled. Prove that the regret can be written as

$$\mathbb{E}[\text{Regret}_T] = \sum_{k=1}^K \mathbb{E}[N_T^{(k)}](\mu^{(k^*)} - \mu^{(k)}).$$

2. [10 points] Consider the ε -greedy policy which at every time step chooses an arm uniformly at random with fixed probability $\epsilon > 0$ and otherwise chooses the greedy arm, i.e. the arm with the highest empirical mean so far. Use the previous part to show that if there exists an arm k with a nonzero gap ($\mu^{(k^*)} > \mu^{(k)}$), then

$$\mathbb{E}[\text{Regret}_T] \geq CT$$

for some constant $C > 0$.

Hint: Start by lower-bounding $\mathbb{E}[N_T^{(k)}]$.

Recall that in lecture, we said that ε -greedy with $\varepsilon_t = \left(\frac{K \ln(t)}{t}\right)^{1/3}$ achieves

$$\text{Regret}_T = \tilde{O}(T^{2/3} K^{1/3}).$$

Although we will not prove this result outright, the remaining parts of this problem are about deriving the choice of $t^{-1/3}$ as the dominating rate in t (i.e., ignoring log factors and K). The following parts do not depend on the preceding two parts of this question.

3. [6 points] Prove the following preliminary bound on the expected regret incurred at time t of an ε -greedy policy using some given ε_t , where R is the event that we are choosing a random arm and R^c is the event that we are exploiting:

$$\mathbb{E} [\mu^{k^*} - \mu^{a_t}] \leq \varepsilon_t + \mathbb{E} [\mu^{k^*} - \mu^{a_t} | R^c]. \quad (1)$$

4. [12 points] Recall from lecture Hoeffding's inequality:

$$P \left(|\hat{\mu}_t^k - \mu^k| \leq \sqrt{\ln(2t/\delta)/2N_t^{(k)}} \right) \geq 1 - \delta$$

Using Hoeffding's inequality with $\delta = \varepsilon_t$, prove the following bound on the regret incurred when we are exploiting at time t of an ε -greedy policy using some given ε_t :

$$\mu^{k^*} - \mu^{a_t} \leq \sqrt{\ln(2Kt/\varepsilon_t)/2} \left(\frac{1}{\sqrt{N_t^{(k^*)}}} + \frac{1}{\sqrt{N_t^{(a_t)}}} \right) \quad \text{w.p.} \geq 1 - \varepsilon_t. \quad (2)$$

5. [10 points] Combining equations (1) and (2), prove the following bound on the expected regret incurred at time t of an ε -greedy policy using some given ε_t :

$$\mathbb{E} [\mu^{k^*} - \mu^{a_t}] \leq 2\varepsilon_t + \sqrt{\ln(2Kt/\varepsilon_t)/2} \mathbb{E} \left[\frac{1}{\sqrt{N_t^{(k^*)}}} + \frac{1}{\sqrt{N_t^{(a_t)}}} \right]. \quad (3)$$

6. [6 points] Prove that for a given arm k ,

$$\mathbb{E} [N_t^{(k)}] \geq \frac{1}{K} \sum_{\tau=0}^{t-1} \varepsilon_\tau. \quad (4)$$

For this final part, the following inequality is good to know (you may use it without proof): for $a \leq 0$,

$$\frac{1}{a+1} ((t+1)^{a+1} - 1) \leq \sum_{\tau=0}^{t-1} (1+\tau)^a \leq \frac{1}{a+1} t^{a+1}. \quad (5)$$

7. [8 points] Now since we're just trying to heuristically figure out ε_t to the leading term in t , we are going to start making some approximations. With the choice $\varepsilon_\tau = (1+\tau)^a$

for some $a \leq 0$ (the “1+” is just to keep us from raising 0 to a power, since τ starts at zero in equation (4)), equations (4) and (5) give

$$\mathbb{E} \left[N_t^{(k)} \right] \geq \frac{1}{K} \frac{1}{a+1} \left((1+t)^{a+1} - 1 \right).$$

Dropping the $\frac{1}{K} \frac{1}{a+1}$ coefficient and the -1 , and dropping the expectation, we get the heuristic inequality

$$N_t^{(k)} \text{ “} \geq \text{” } (1+t)^{a+1},$$

where the quotations around the inequality sign are a reminder that this is not a rigorous inequality. If we then plug this inequality and $\varepsilon_\tau = (1+\tau)^a$ into equation (3) and drop all constant and logarithmic coefficients, and finally sum over T to get the total regret at time T , we get

$$\mathbb{E}[\text{Regret}_T] \text{ “} \leq \text{” } \sum_{t=0}^{T-1} \left((1+t)^a + \frac{1}{\sqrt{(1+t)^{a+1}}} \right).$$

Show that the right-hand side of the above equation is $\tilde{O}(T^{2/3})$ for the choice of $a = -1/3$.

Hint: Recall that for a nonincreasing function, the Riemann sum using the right endpoint is a lower bound on the integral.